PSX Market Intelligence & Portfolio Optimization System

1. Introduction

The Pakistan Stock Exchange (PSX) is the primary platform for equity trading in Pakistan and plays a crucial role in capital formation, investment activities, and economic growth. With the increasing availability of historical financial data, modern data analytics techniques can be applied to analyze market behavior, evaluate trading strategies, and support data-driven investment decisions.

This project develops a **comprehensive PSX Market Intelligence & Portfolio Optimization System** using **Python and Power BI**. The system follows a complete **end-to-end data science pipeline**, starting from raw data collection to advanced portfolio optimization and interactive visualization.

The project simulates the **real-world workflow of a data analyst or quantitative analyst** working in investment research, trading desks, or fintech companies, where data engineering, analysis, modeling, and business interpretation are equally important.

2. Objectives of the Project

The main objectives of this project are:

- To collect historical stock and index data from authentic financial sources
- To design an automated ETL (Extract, Transform, Load) pipeline using Python
- To perform descriptive and technical analysis of selected PSX-listed companies
- To evaluate rule-based trading strategies using back-testing and performance metrics
- To apply Modern Portfolio Theory for optimal asset allocation
- To present analytical insights through professional, interactive Power BI dashboard.

3. Data Collection & Dataset Description

3.1 Data Sources

Multiple data sources were used to ensure completeness and realism:

- **Company Stock Data (OHLCV)**
  Downloaded programmatically using the **Python yfinance library** for six PSX-listed companies:
  - **Banking:** HBL, MCB
  - **Cement:** DGKC, LUCK
  - **Oil & Gas:** OGDC, PPL
- **KSE-100 Index Data**
  Downloaded manually from **Investing.com** due to limited API availability for the PSX index.

3.2 Time Period

- Daily historical data from **2020 to 2025**

3.3 Supporting Datasets

- Sector classification file
- Technical indicators (Moving Averages, RSI)
- Trading strategy results (MA & RSI strategies)
- Portfolio simulation results (Monte Carlo portfolios)

4. Data Preprocessing & ETL Pipeline

After data collection, a **fully automated ETL pipeline** was developed in Python to ensure consistency and reproducibility.

Key ETL Steps:

- Standardized date formats across all datasets
- Converted numerical columns to appropriate data types

- Removed missing values and duplicate records
- Ensured **one observation per stock per trading day**
- Merged company-level data with:
    - Sector information
    - KSE-100 index data
- Calculated:
    - Daily returns
    - Weekly returns
    - Monthly returns
    - Annual returns

The cleaned and transformed data was merged into a **single master dataset**, which served as the foundation for all subsequent analysis, modeling, and visualization tasks.

This automated ETL process reflects **industry-level data engineering practices**.

5. Descriptive Analytics & Visualization Tasks

5.1 Descriptive Analytics

Using Python (Pandas & NumPy), the following descriptive metrics were computed:

- PSX daily, weekly, monthly, and annual returns
- Sector-wise performance comparison
- Market-wide return behavior relative to the KSE-100 index

These calculations provided a quantitative understanding of market trends before applying advanced techniques.

5.2 Visualization Tasks (Python)

Before Power BI, exploratory visualizations were created in Python:

- Price trend line charts for all companies
- Candlestick charts to analyze daily price behavior
- Sector-wise performance plots

These visualizations helped validate trends and patterns at an early stage.

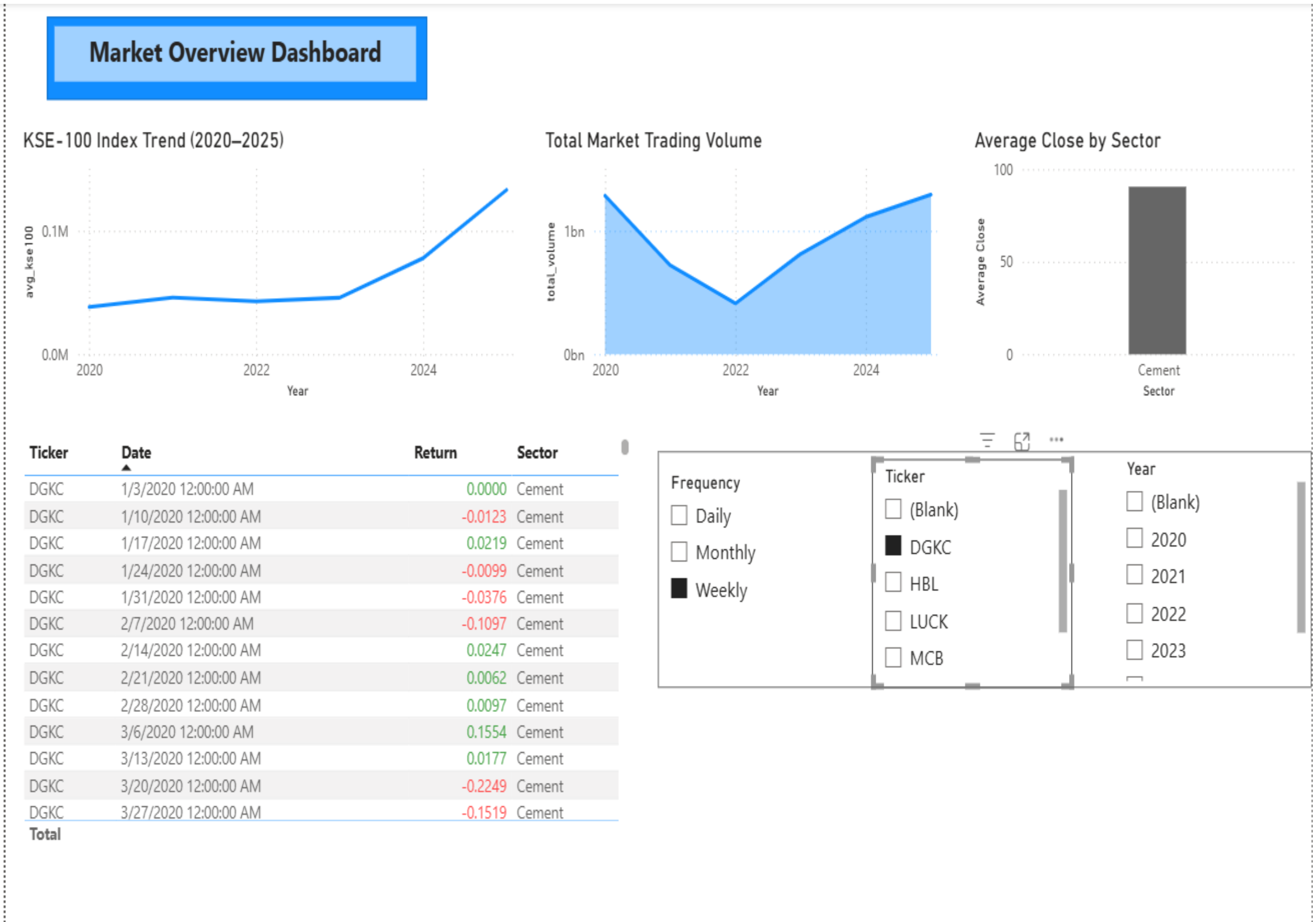
6. Market Overview Analysis — Power BI Dashboard (Page 1)

This dashboard provides a **macro-level snapshot of the PSX market**.

Key Insights:

- The **KSE-100 Index** shows a strong upward trend after 2023, indicating market recovery and renewed investor confidence.
- Market trading volume increases significantly during bullish periods, reflecting higher participation.
- Banking sector stocks exhibit higher average prices, indicating relative stability and stronger fundamentals.
- Cement and Oil & Gas sectors display higher volatility, suggesting cyclical and commodity-driven behavior.

This page is designed for **investors and decision-makers** who need a high-level market overview.



7. Company-Level Technical Analysis — Power BI Dashboard (Page 2)

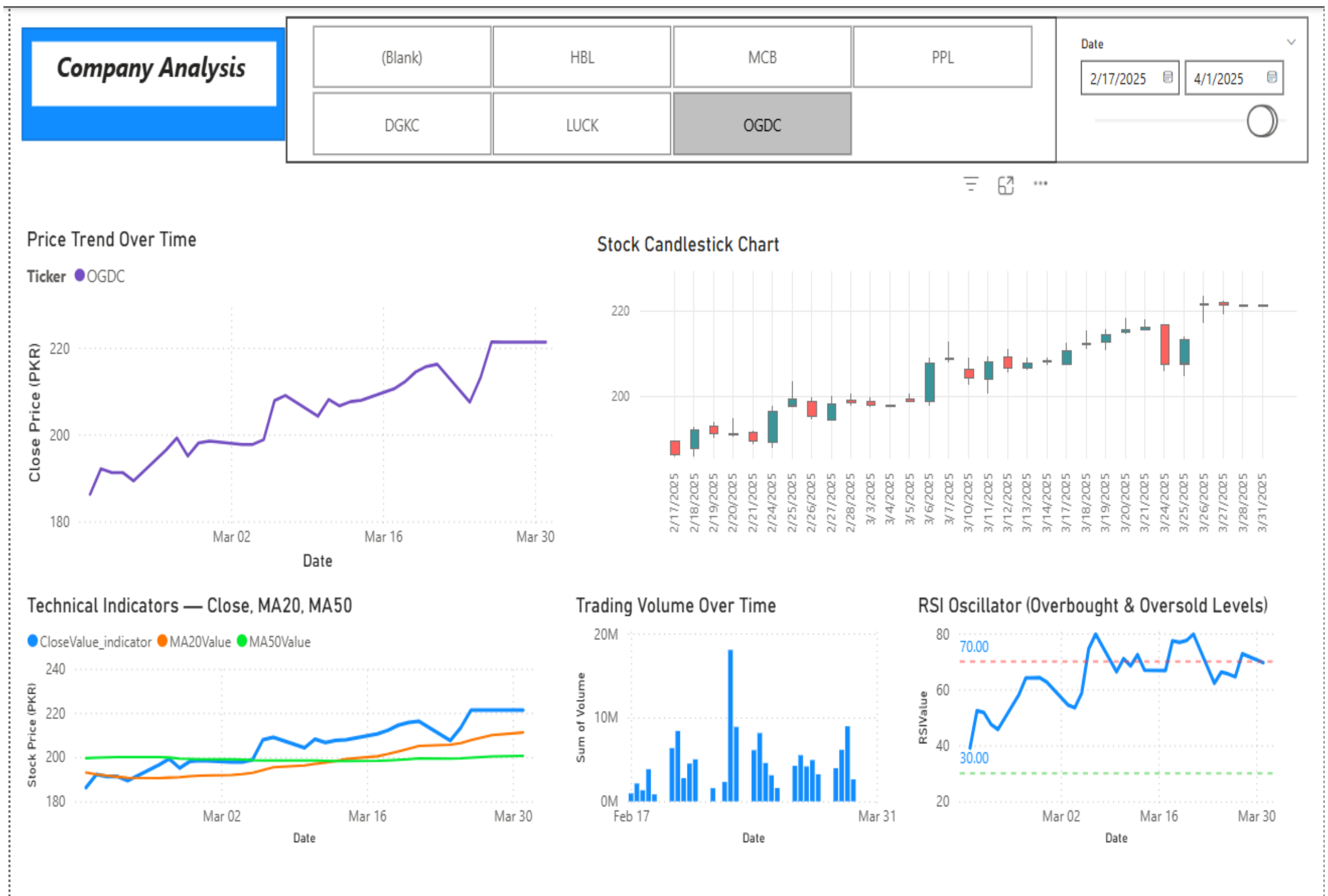
This dashboard replicates a **professional stock analysis terminal**.

Techniques Applied:

- Price trend analysis
- Moving averages (MA20, MA50) for trend identification
- Relative Strength Index (RSI) for momentum analysis
- Candlestick charts for daily price action
- Volume analysis for trade confirmation

Key Observations:

- Moving averages help identify medium-term trend direction.
- RSI values above 70 indicate overbought conditions, while values below 30 indicate oversold conditions.
- Candlestick patterns reveal short-term market sentiment and momentum shifts.



8. Trading Strategy Performance Analysis — Power BI Dashboard (Page 3)

Two quantitative trading strategies were implemented and evaluated:

- **Moving Average (MA) Crossover Strategy**
- **RSI-Based Mean Reversion Strategy**

Each strategy was compared against a **Buy & Hold benchmark.**

Performance Metrics Used:

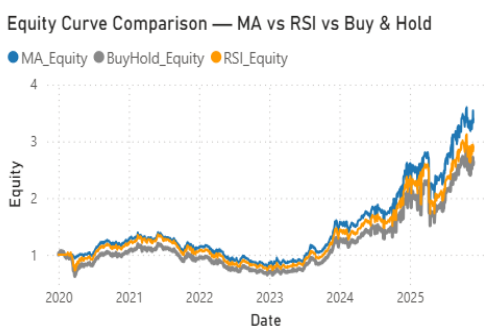
- Total Return
- Sharpe Ratio (risk-adjusted performance)
- Maximum Drawdown

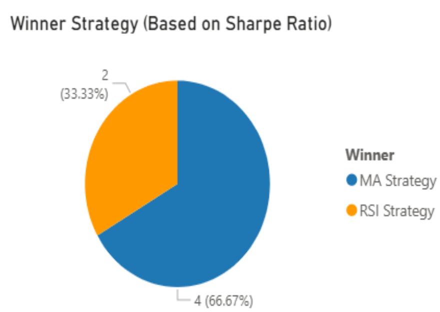
Key Findings:

- MA strategy outperformed RSI strategy for most stocks.
- RSI strategy performed better during sideways or range-bound markets.
- Buy & Hold underperformed during high-volatility periods.
- MA strategy delivered higher risk-adjusted returns but occasionally experienced larger drawdowns.

This analysis demonstrates how **rule-based quantitative strategies can outperform passive investing** under specific market conditions.

Trading Strategy Performance





Ticker
- (Blank)
- DGKC
- HBL
- LUCK
- MCB
- OGDC
- PPL

Ticker	Sum of MA_Total_Return	Sum of RSI_Total_Return	Sum of MA_Sharpe	Sum of RSI_Sharpe	Sum of MA_Max_Drawdown	Sum of RSI_Max_Drawdown
LUCK	0.00	0.26	0.00	0.02	-0.82	-0.82
PPL	1.89	1.31	0.56	0.41	-0.47	-0.53
HBL	3.81	2.10	1.05	0.68	-0.51	-0.58
OGDC	2.79	2.21	0.78	0.63	-0.31	-0.44
DGKC	1.86	2.46	0.50	0.59	-0.71	-0.71
MCB	4.06	3.11	1.39	1.09	-0.26	-0.36
Total	**14.39**	**11.46**	**4.28**	**3.42**	**-3.08**	**-3.45**

9. Portfolio Optimization — Power BI Dashboard (Page 4)

Modern Portfolio Theory was applied using **Monte Carlo simulation**.

Methodology:

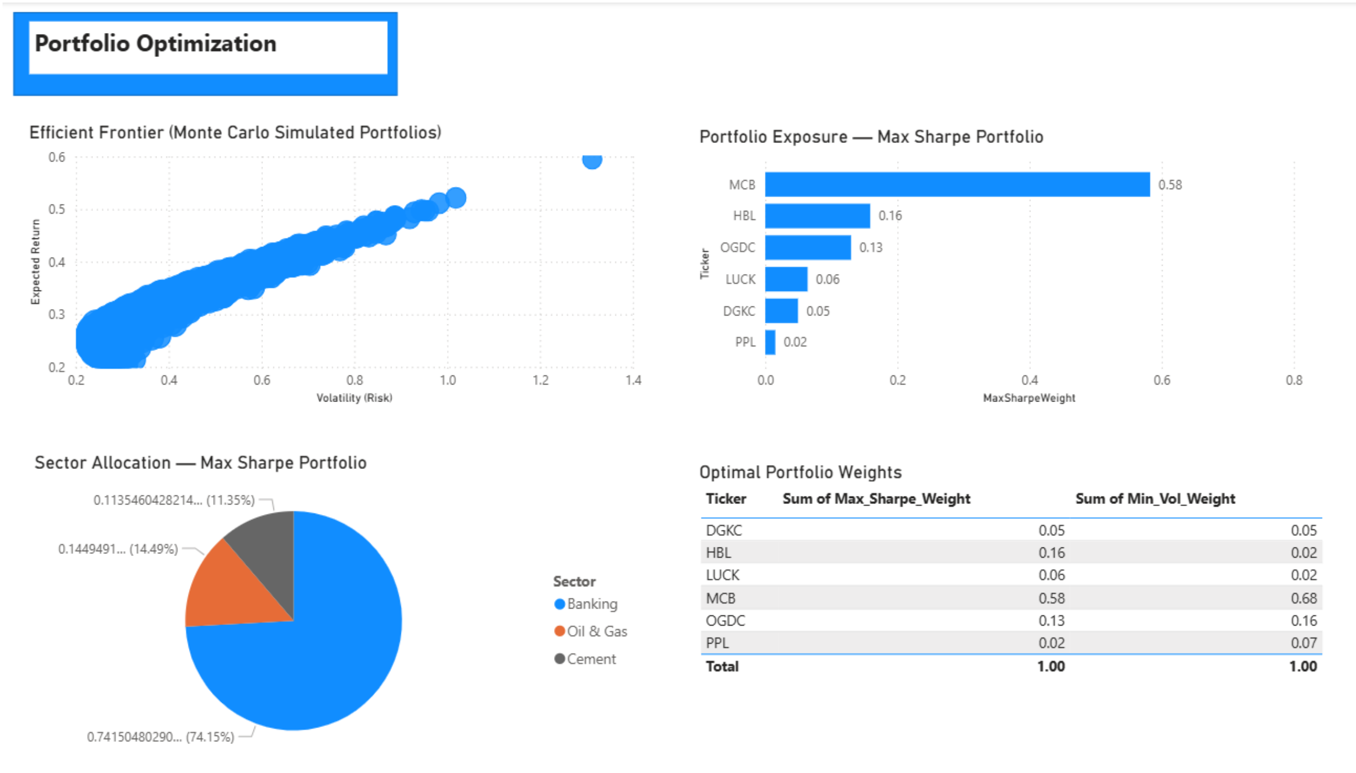
- Thousands of random portfolios were simulated.
- Expected return, volatility, and Sharpe Ratio were calculated.
- The **Efficient Frontier** was constructed.

Optimal Portfolios Identified:

- **Maximum Sharpe Ratio Portfolio:** Best risk-adjusted return
- **Minimum Volatility Portfolio:** Lowest overall risk

Insights:

- Banking stocks received the highest weights due to stable returns.
- Cement and Oil & Gas stocks received smaller allocations due to higher volatility.
- The optimal portfolio is **moderately aggressive**, suitable for medium-term investors.



Ticker	Sum of Max_Sharpe_Weight	Sum of Min_Vol_Weight
DGKC	0.05	0.05
HBL	0.16	0.02
LUCK	0.06	0.02
MCB	0.58	0.68
OGDC	0.13	0.16
PPL	0.02	0.07
Total	**1.00**	**1.00**

10. Tools & Technologies Used

- **Python:** Pandas, NumPy, Matplotlib, Scikit-learn
- **Power BI:** Interactive dashboards and BI modeling
- **Excel / CSV:** Data exchange and validation
- **Monte Carlo Simulation:** Portfolio optimization
- **yfinance:** Automated stock data collection

11. Conclusion

This project demonstrates how data science and analytics techniques can be applied to real-world financial market data. By integrating **data collection, ETL, descriptive analytics, technical analysis, trading strategy evaluation, and portfolio optimization**, the system delivers actionable investment insights.

The project reflects **industry-level analytical practices** and highlights the importance of combining technical analysis with business interpretation.

12. Limitations & Future Work

Limitations:

- Analysis is based on historical data only
- Transaction costs and taxes are not included
- Limited number of companies considered

Future Enhancements:

- Integration of macroeconomic indicators (interest rates, inflation)
- Machine learning-based price forecasting models
- Real-time data integration using APIs
- Expansion to sector-wide and index-level portfolios